

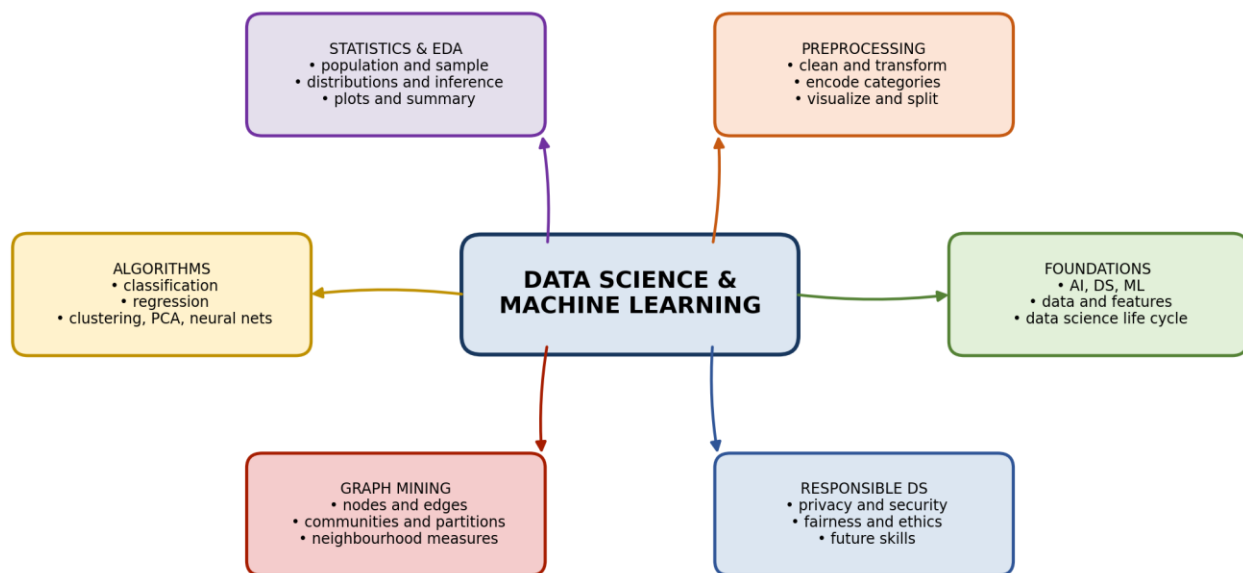
DATA SCIENCE AND MACHINE LEARNING

Basic Concepts through Diagrams, Mind Maps and Visual Revision Notes

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Master Mind Map of the Subject



Units 1–5 | Beginner Friendly | Classroom and Examination Support

How to Use These Visual Notes

These notes are designed to build the mental picture first and add definitions, formulas and code only after the idea is clear. Study each topic in three passes:

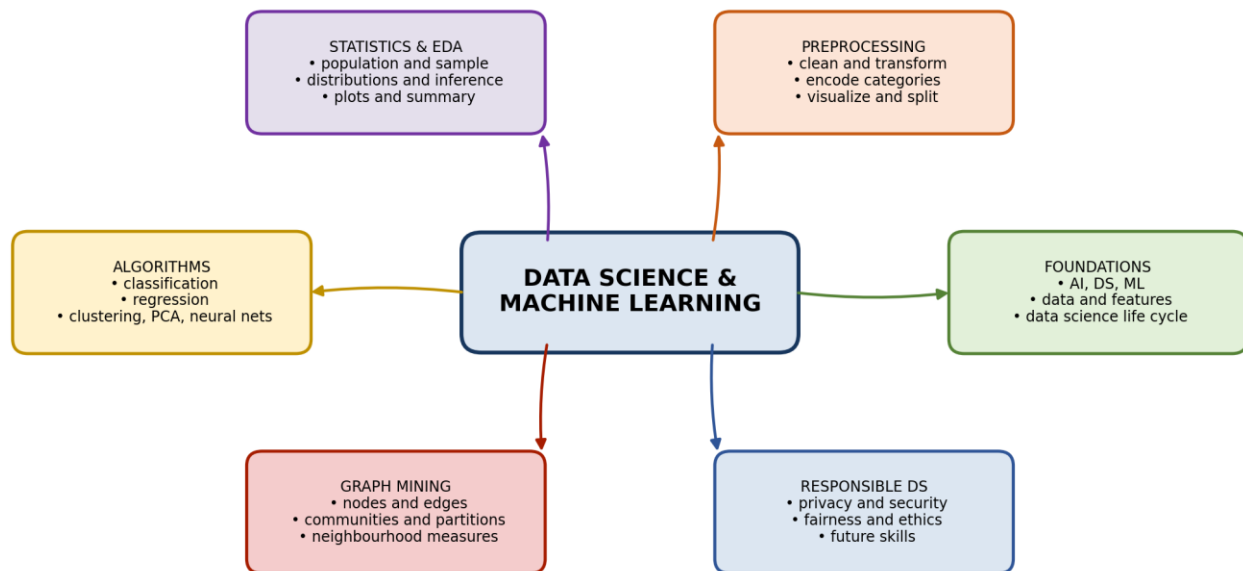
Pass	What to do	Result
1. See	Observe the diagram or mind map and name every part.	You understand the overall structure.
2. Say	Explain the figure in your own words without reading.	You convert the picture into knowledge.
3. Solve	Attempt the checkpoint, formula or small example.	You confirm that the concept is usable.

MEMORY RULE: Golden learning rule

Do not memorise an algorithm as code. First identify its input, output, learning idea, important setting and evaluation measure.

Master Subject Map

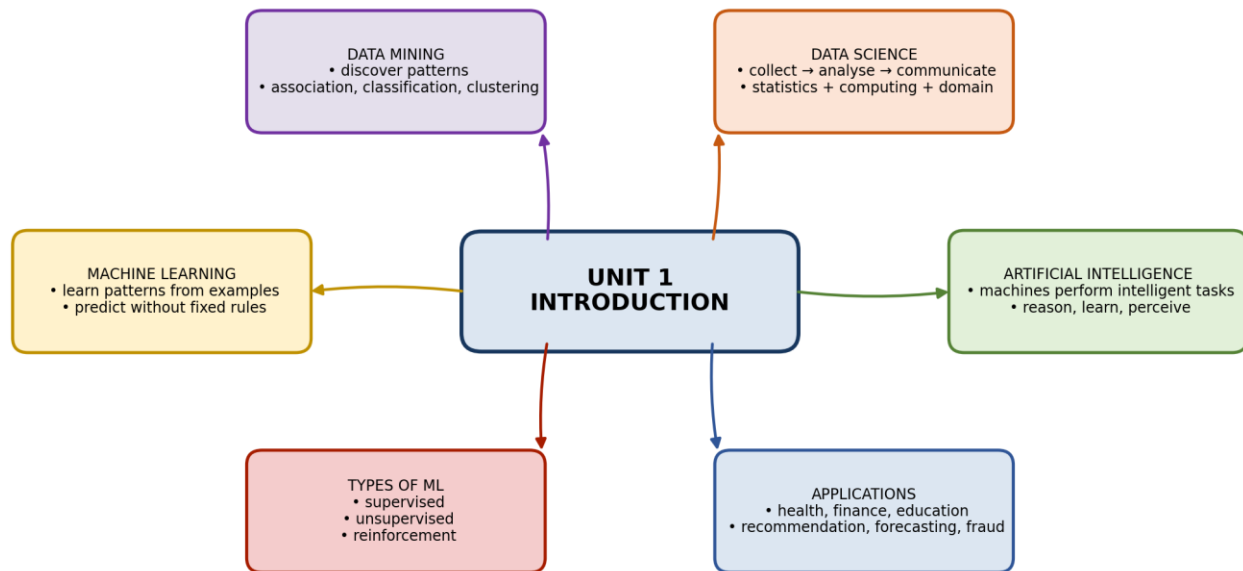
Master Mind Map of the Subject



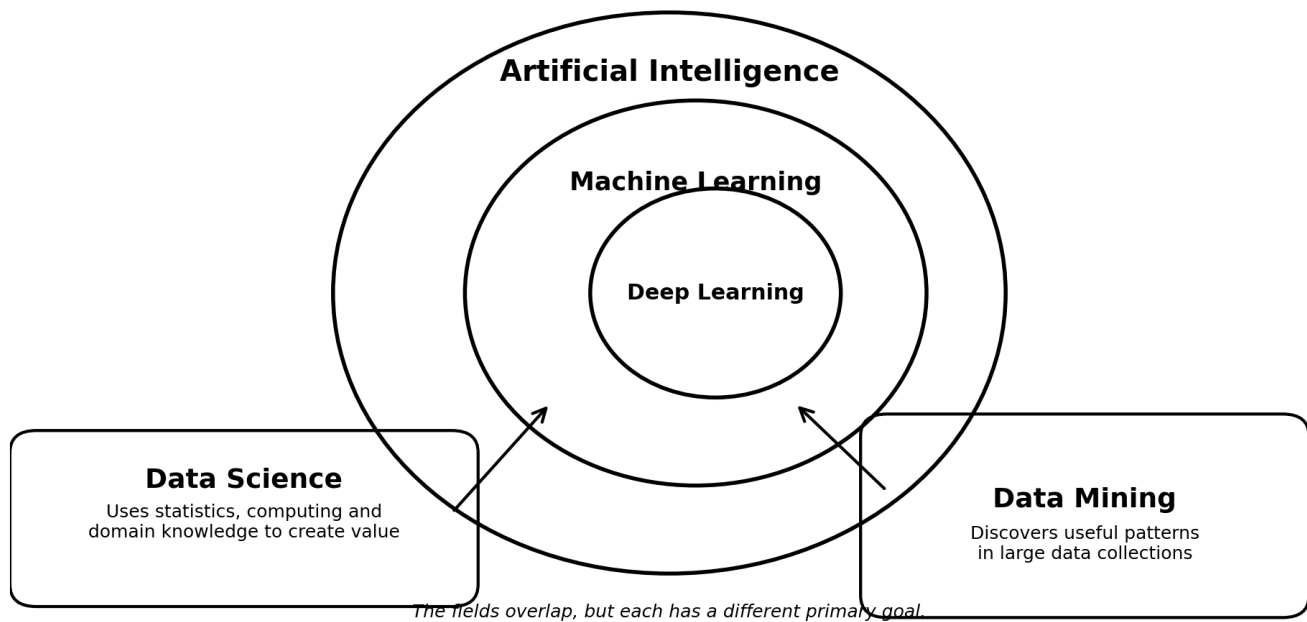
The five units connect data, statistics, algorithms, graphs and responsible practice.

UNIT 1 — Foundations of AI, Data Science and Machine Learning

Unit 1 Mind Map



1. The Big Relationship



Artificial Intelligence

The broad goal of building systems that perform tasks requiring intelligence, such as reasoning, perception and decision-making.

Machine Learning

A method of achieving AI by learning patterns from examples instead of writing every rule manually.

Data Science

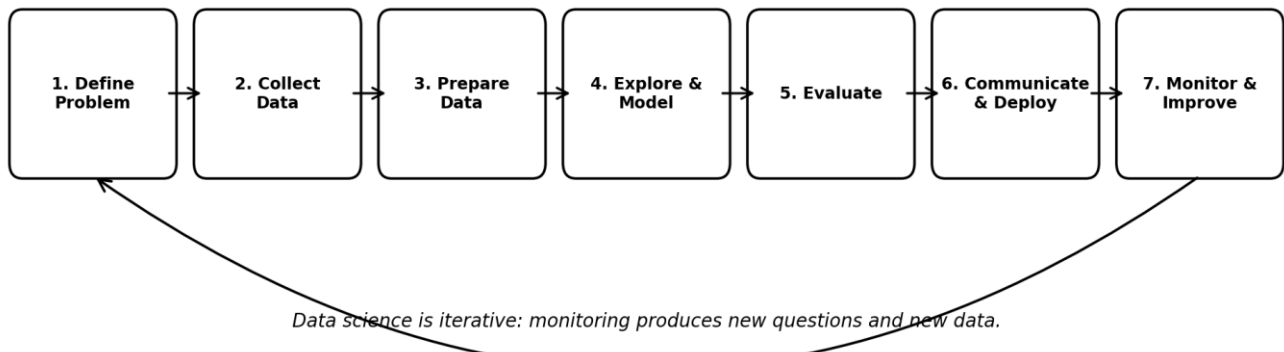
A complete process that uses data, statistics, computing and domain knowledge to answer questions and create value.

Data Mining

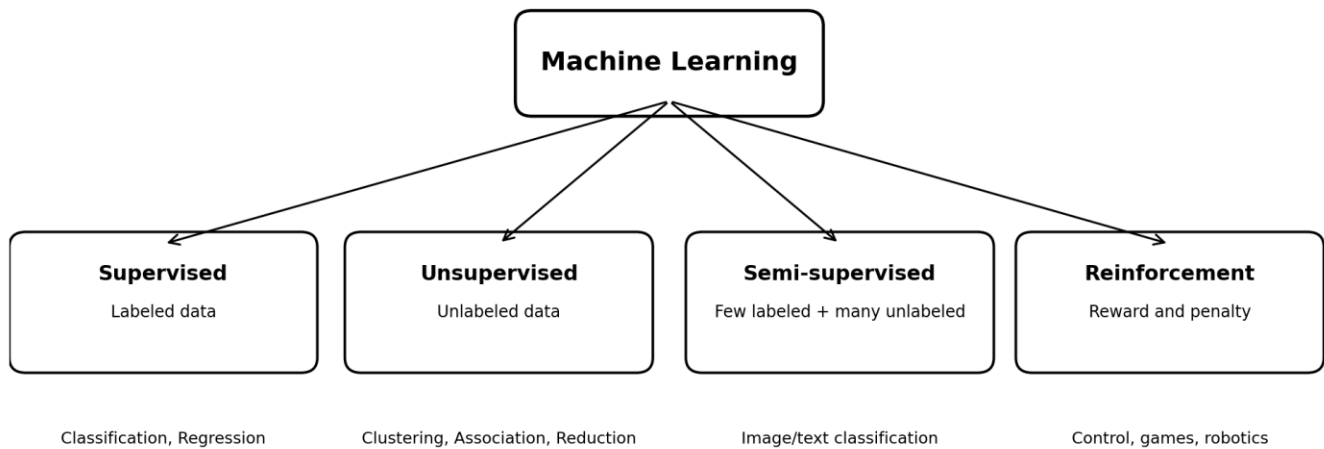
The activity of discovering useful patterns, relationships or groups inside large datasets.

ANALOGY: Simple analogy

A hospital wants to reduce patient waiting time. Data science defines the problem and analyses records; data mining finds patterns; machine learning predicts waiting time; AI may use the prediction to schedule resources.

2. Data Science Life Cycle**Typical Data Science Life Cycle**

- **Ask a question:** What decision must improve?
- **Collect data:** Gather relevant, lawful and reliable information.
- **Prepare data:** Clean, combine and transform it.
- **Explore and model:** Understand patterns and build suitable models.
- **Communicate and deploy:** Explain results and use them in a real system.
- **Monitor:** Check accuracy, fairness and changes over time.

3. Types of Machine Learning

Choose the learning type by asking: What feedback or labels are available?

Type	Learning signal	Typical question	Example
Supervised	Correct output is known	What will the output be?	Predict pass/fail or house price

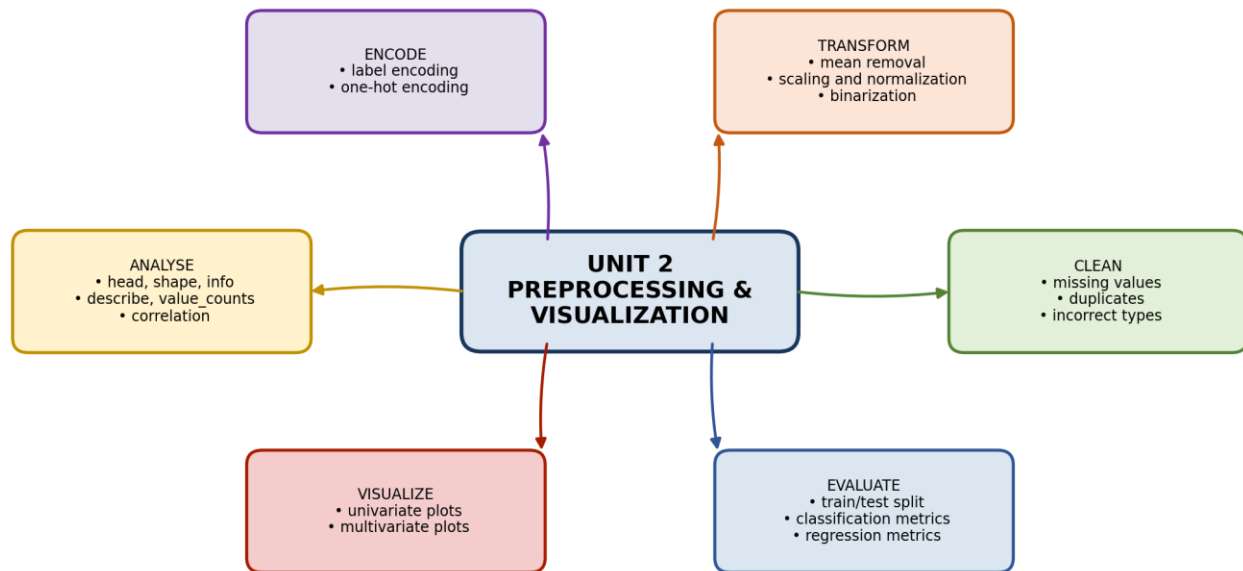
Type	Learning signal	Typical question	Example
Unsupervised	No correct output labels	What structure exists?	Group customers by behaviour
Reinforcement	Reward or penalty	Which action gives long-term reward?	Game-playing agent or robot control

THINK: Checkpoint

Can you explain why customer grouping is unsupervised but predicting whether a customer will leave is supervised?

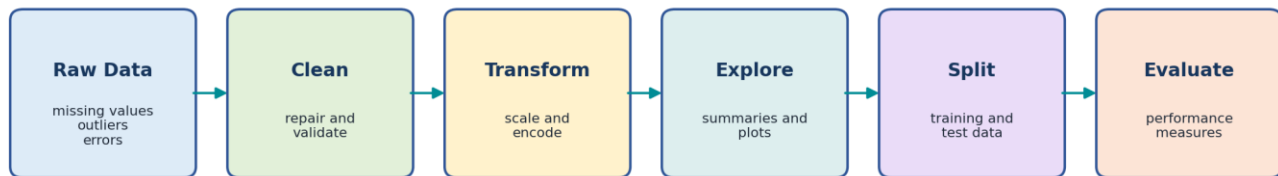
UNIT 2 — Data Preprocessing, Analysis and Visualization

Unit 2 Mind Map



1. Why Preprocessing Comes Before Learning

UNIT 02 WORKFLOW: FROM RAW DATA TO TRUSTWORTHY EVALUATION

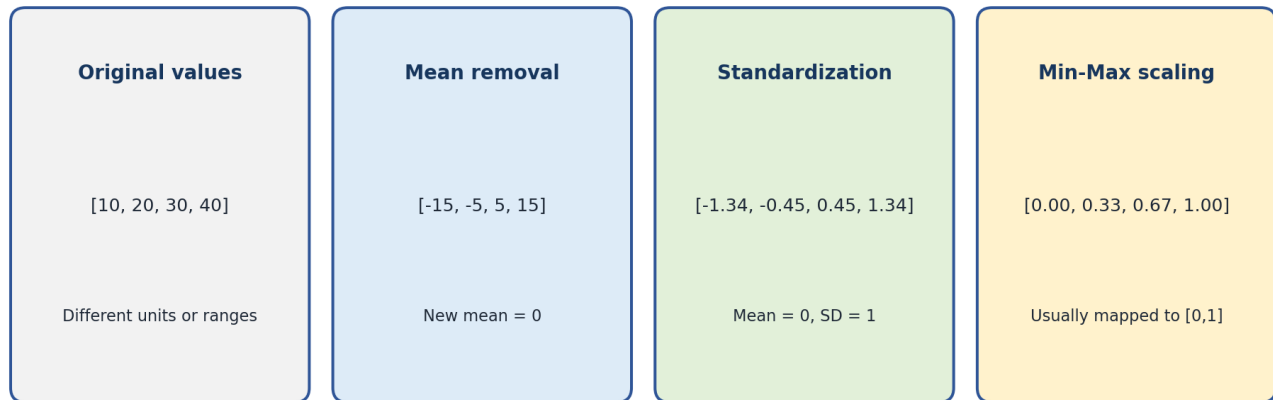


WARNING: Garbage in, garbage out

A powerful algorithm cannot repair incorrect labels, inconsistent units, missing values or data leakage automatically. Reliable learning begins with reliable input.

2. Scaling Concepts at a Glance

HOW NUMERIC PREPROCESSING CHANGES THE SAME FEATURE



Technique	Main idea	Typical formula/behaviour	When useful
Mean removal	Shift centre to zero	$x - \text{mean}$	Data should be centred
Standardization	Mean 0 and standard deviation 1	$(x - \text{mean})/\text{standard deviation}$	SVM, k-NN, regression
Min-max scaling	Map values to a fixed interval	Usually 0 to 1	Bounded inputs, neural networks
Normalization	Scale each row/vector to unit length	Length becomes 1	Text vectors, cosine similarity
Binarization	Convert value to 0 or 1	Compare with threshold	Presence/absence decisions

3. Encoding Categorical Data

CATEGORICAL ENCODING EXAMPLE

City	Label code	Kanpur	Delhi	Lucknow
Kanpur	0	1	0	0
Delhi	1	0	1	0
Lucknow	2	0	0	1

Label encoding: one integer per category

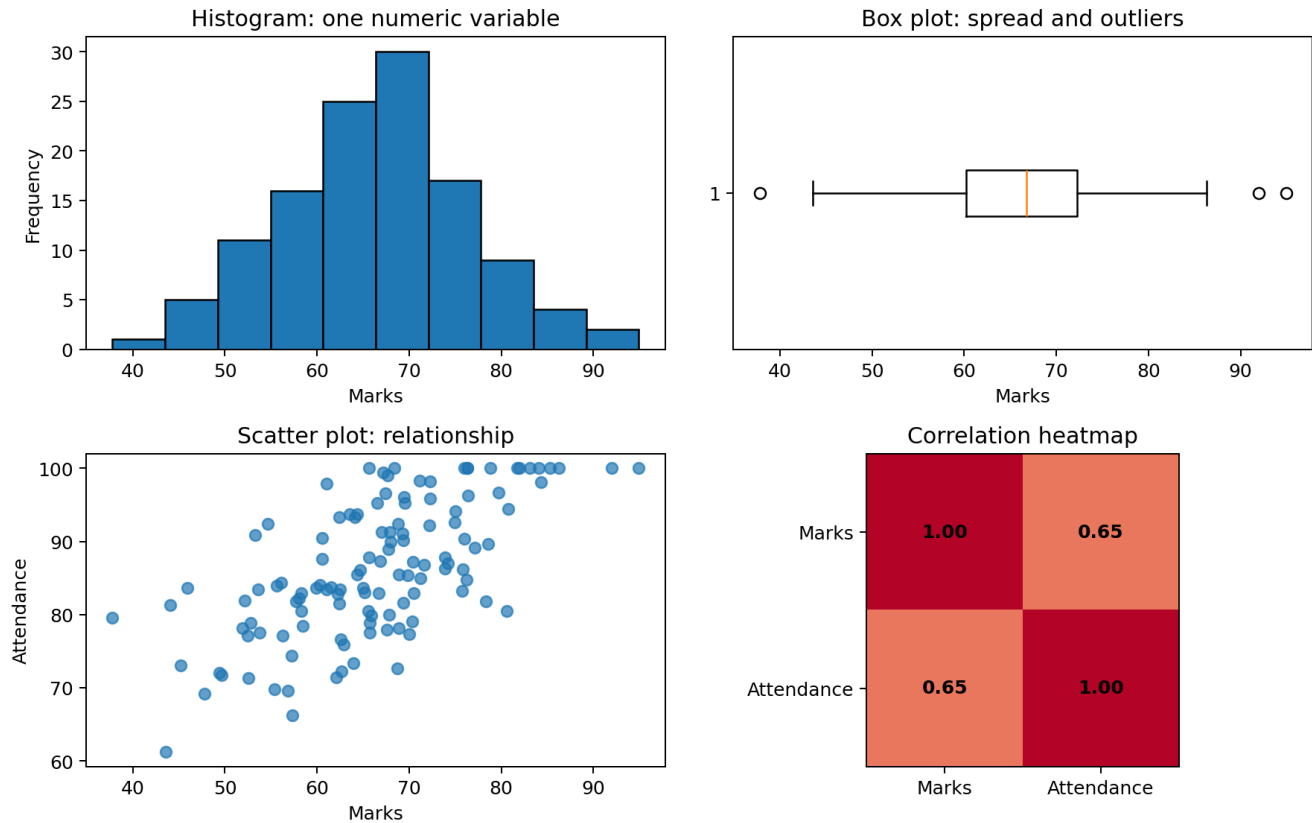
One-hot encoding: one binary column per category

DO NOT CONFUSE: Common confusion

Label encoding numbers are category codes, not measured quantities. One-hot encoding is safer for nominal input features because it does not create a false order.

4. Analysis and Visualization

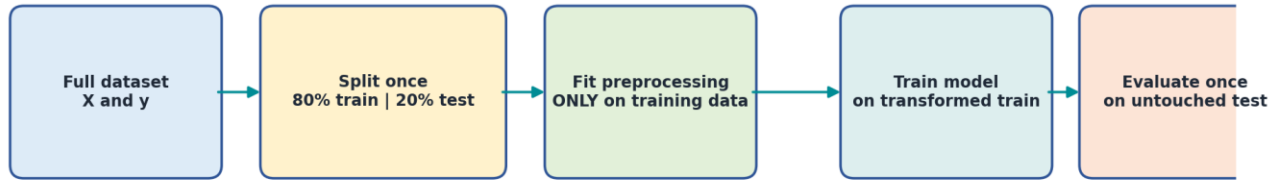
UNIVARIATE AND MULTIVARIATE VISUALIZATION



Question	Recommended visual
How is one numeric variable distributed?	Histogram or box plot
How frequent is each category?	Bar chart
How do two numeric variables move together?	Scatter plot
How do several numeric variables relate?	Pair plot or correlation heatmap
How does a value change with time?	Line chart

5. Training Data, Test Data and Fair Evaluation

CORRECT TRAIN-TEST WORKFLOW

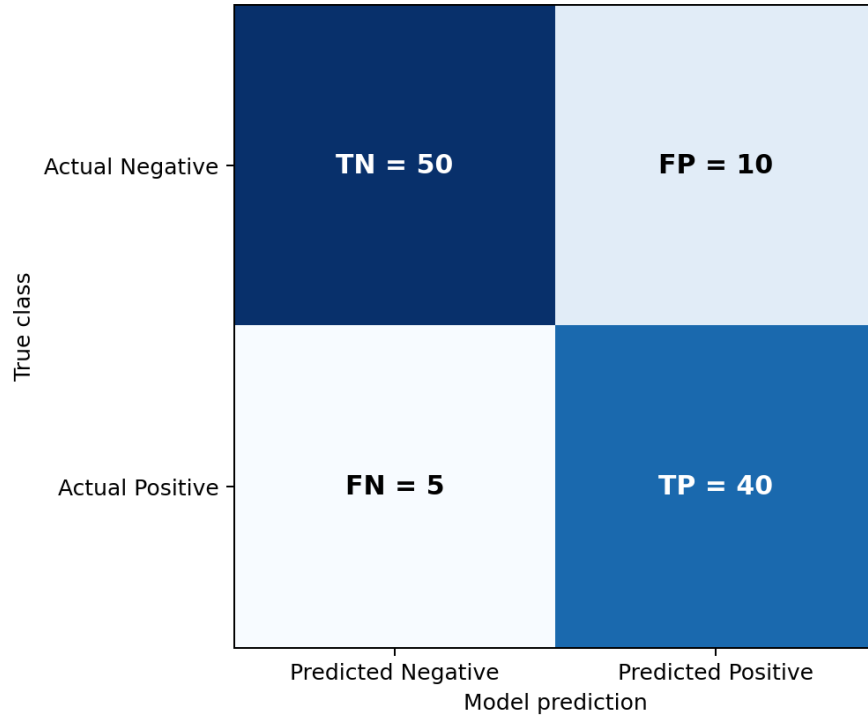


Never calculate scaling parameters from the complete dataset: that leaks test information.

Training set Used to fit model parameters and learn patterns.	Validation set Used to compare settings and select a model.
Test set Used once at the end to estimate performance on unseen data.	Data leakage Information from the test/future data enters training, producing unrealistically high performance.

6. Classification Performance

Confusion Matrix Example



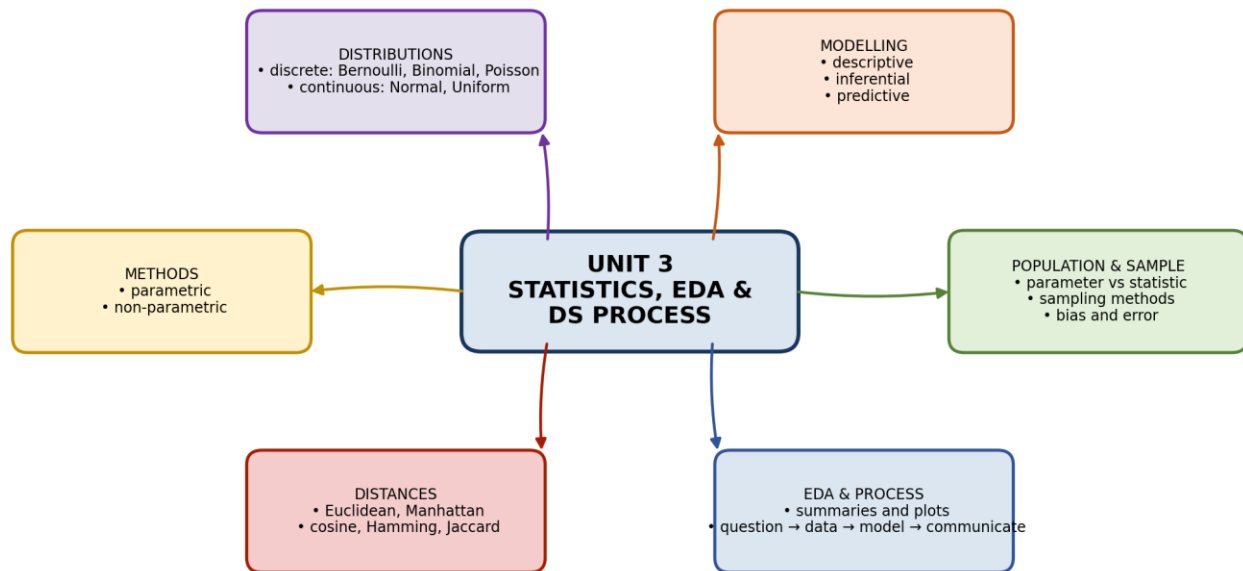
Measure	Meaning	Best focus
Accuracy	Fraction of all predictions that are correct	Balanced classes and equal error cost
Precision	Of predicted positives, how many are truly positive?	False alarms are costly
Recall	Of actual positives, how many were found?	Missing positives is costly
F1-score	Balance between precision and recall	Need a single balanced score

REMEMBER: Memory aid

Precision looks at the model's positive predictions. Recall looks at the real positive cases.

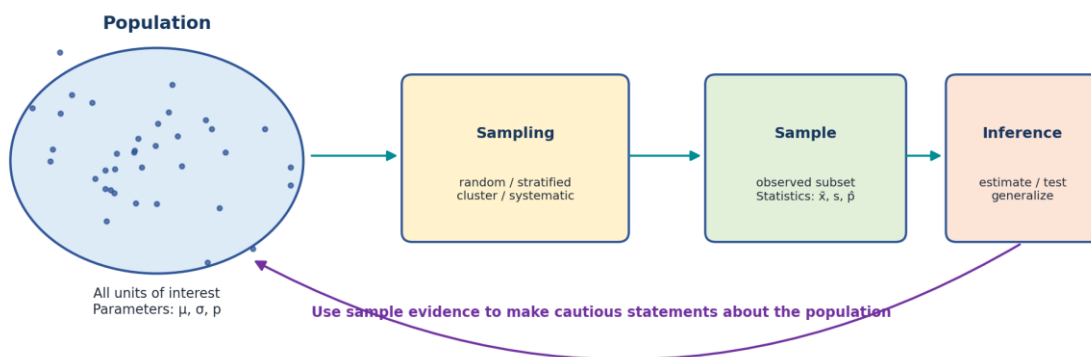
UNIT 3 — Statistical Inference, EDA and the Data Science Process

Unit 3 Mind Map



1. Population, Sample, Parameter and Statistic

FROM POPULATION TO SAMPLE AND BACK TO INFERENCE



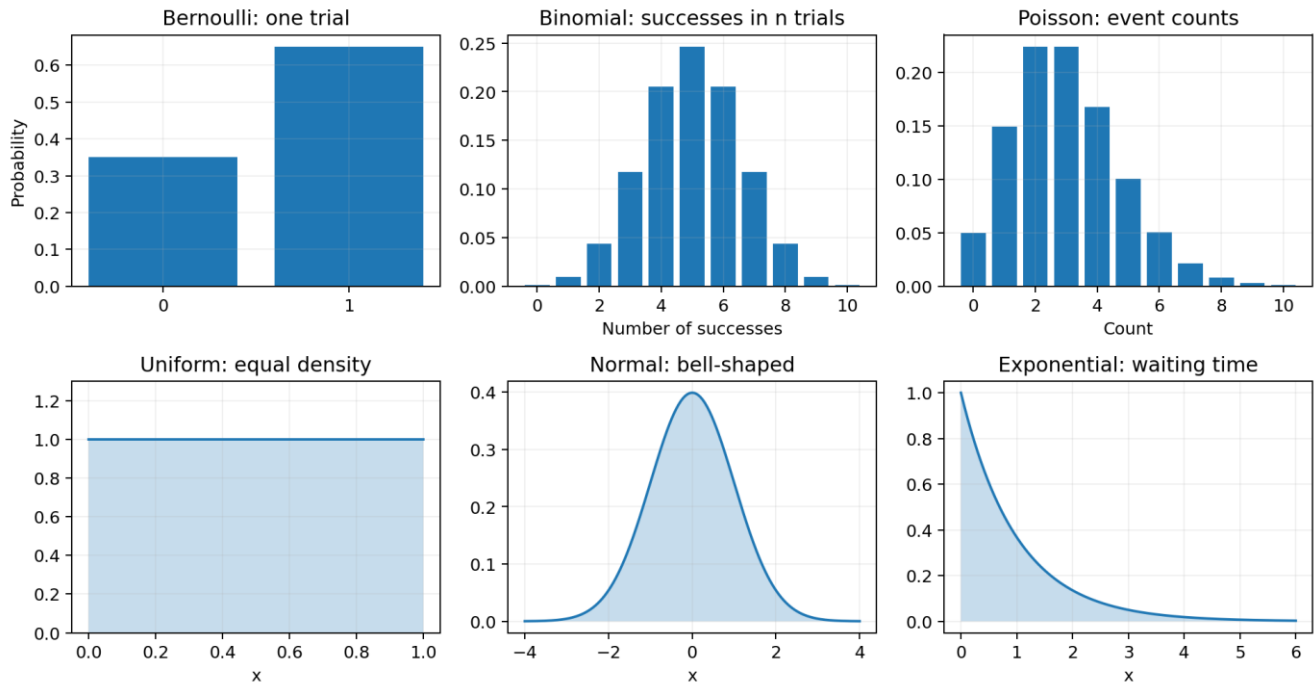
Term	Meaning	Example
Population	The complete group of interest	All students in a university
Sample	A smaller observed subset	300 selected students
Parameter	True but often unknown population quantity	Average attendance of all students
Statistic	Quantity calculated from the sample	Average attendance of 300 students

CORE IDEA: Inference

Statistical inference uses a sample statistic, together with uncertainty, to make a careful statement about a population parameter.

2. Probability Distributions

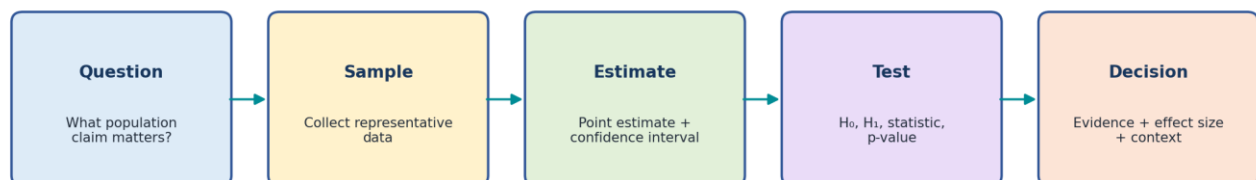
COMMON DISCRETE AND CONTINUOUS PROBABILITY DISTRIBUTIONS



Distribution	Type	Typical situation
Bernoulli	Discrete	One yes/no trial
Binomial	Discrete	Number of successes in fixed trials
Poisson	Discrete	Event count in time or space
Uniform	Continuous	All values in an interval equally likely
Normal	Continuous	Symmetric measurements around a mean
Exponential	Continuous	Waiting time between Poisson-type events

3. Statistical Inference Workflow

STATISTICAL INFERENCE WORKFLOW



Inference quantifies uncertainty; it does not eliminate uncertainty.

Key Terms in Hypothesis Testing

Term	Student-friendly meaning
Null hypothesis (H0)	The default claim: no effect, difference or relationship.

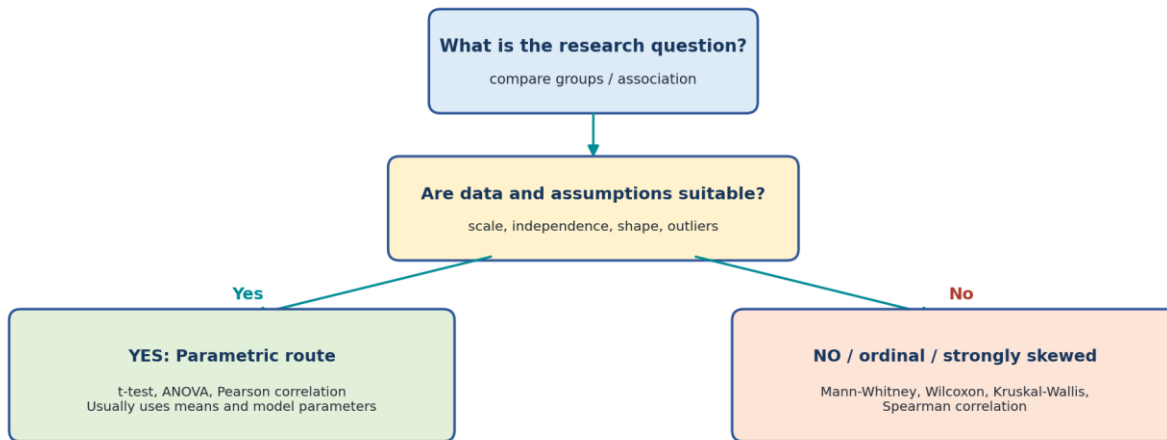
Term	Student-friendly meaning
Alternative hypothesis (H1)	The claim supported when evidence against H0 is strong.
p-value	How surprising the observed result would be if H0 were true.
Significance level	A pre-decided evidence threshold, often 0.05.
Confidence interval	A range of plausible values for a population quantity.

GOOD PRACTICE: Practical interpretation

Report the estimated effect and uncertainty, not only whether a p-value crosses a threshold.

4. Parametric versus Non-Parametric Methods

CHOOSING BETWEEN PARAMETRIC AND NON-PARAMETRIC METHODS

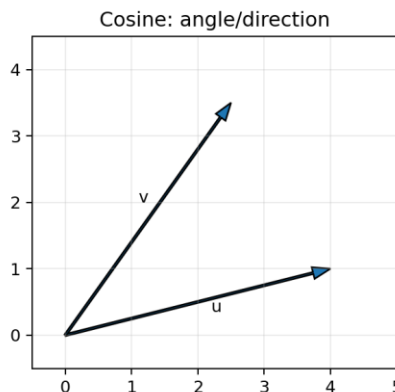
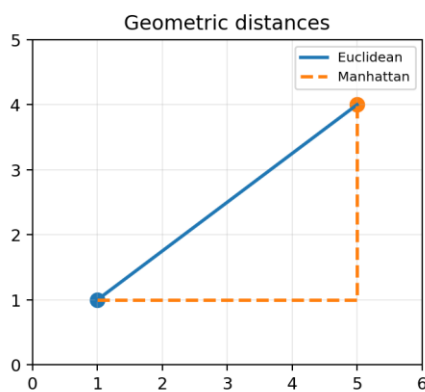


Do not choose a test only from habit. Match the method to the question, measurement scale and assumptions.

Feature	Parametric	Non-parametric
Assumptions	Specific distribution/form assumptions	Fewer or weaker assumptions
Uses	Means and model parameters	Ranks, signs or empirical structure
Examples	t-test, ANOVA, Pearson correlation	Mann-Whitney, Wilcoxon, Spearman
Advantage	Often more powerful when assumptions hold	Robust for ordinal/skewed/non-normal data

5. Distance Metrics

DISTANCE METRICS MEASURE DIFFERENT NOTIONS OF "CLOSENESS"



Binary/set distances

Hamming

fraction or count of positions that differ

Jaccard

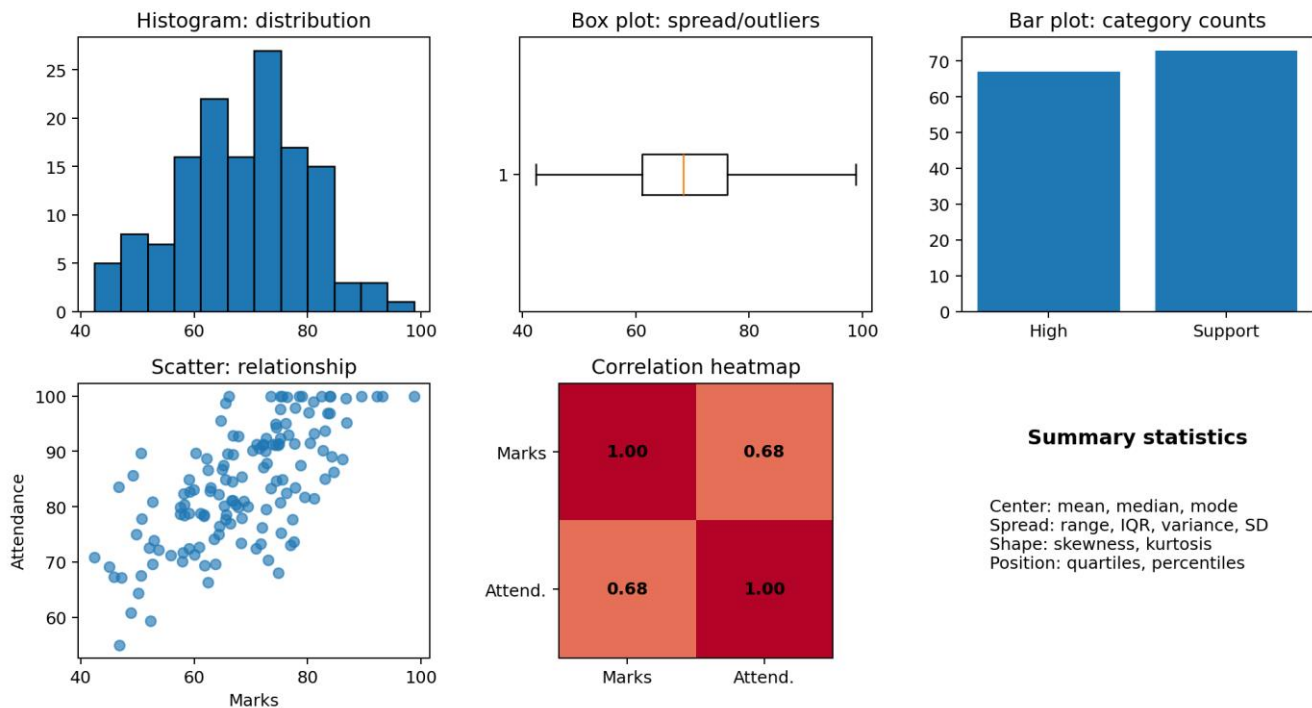
1 - intersection / union for presence sets

SELECTION RULE: Choose distance to match meaning

Euclidean suits straight-line numeric distance; Manhattan sums coordinate differences; cosine compares direction; Hamming counts mismatches; Jaccard compares shared items.

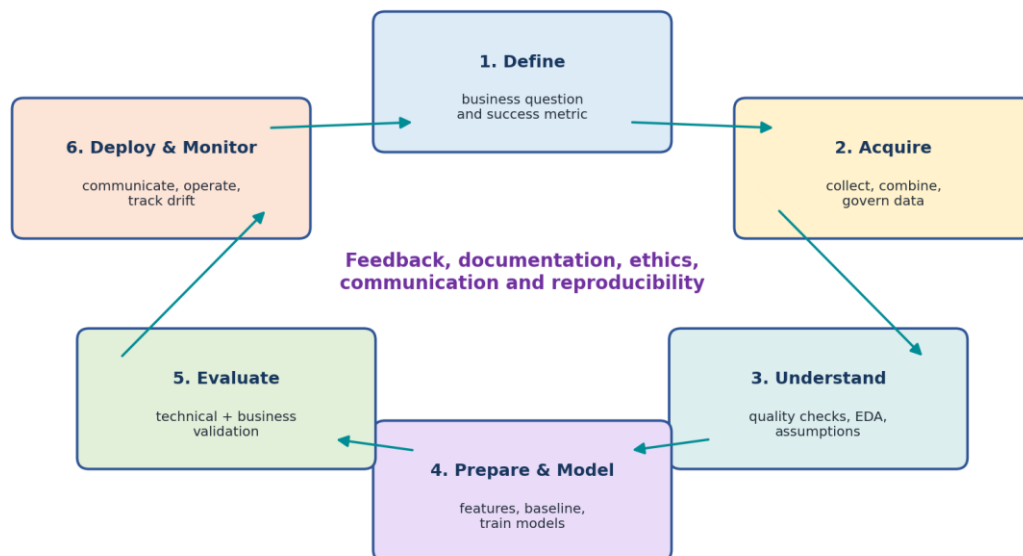
6. Exploratory Data Analysis

BASIC EDA TOOLBOX: NUMBERS AND VISUALS WORK TOGETHER



EDA asks What patterns, errors, relationships and unusual cases exist?	EDA uses Summary statistics, distributions, plots, grouping and comparisons.
EDA does not prove A visible pattern is not automatically causal or statistically reliable.	EDA produces Better questions, better features, suitable models and honest limitations.

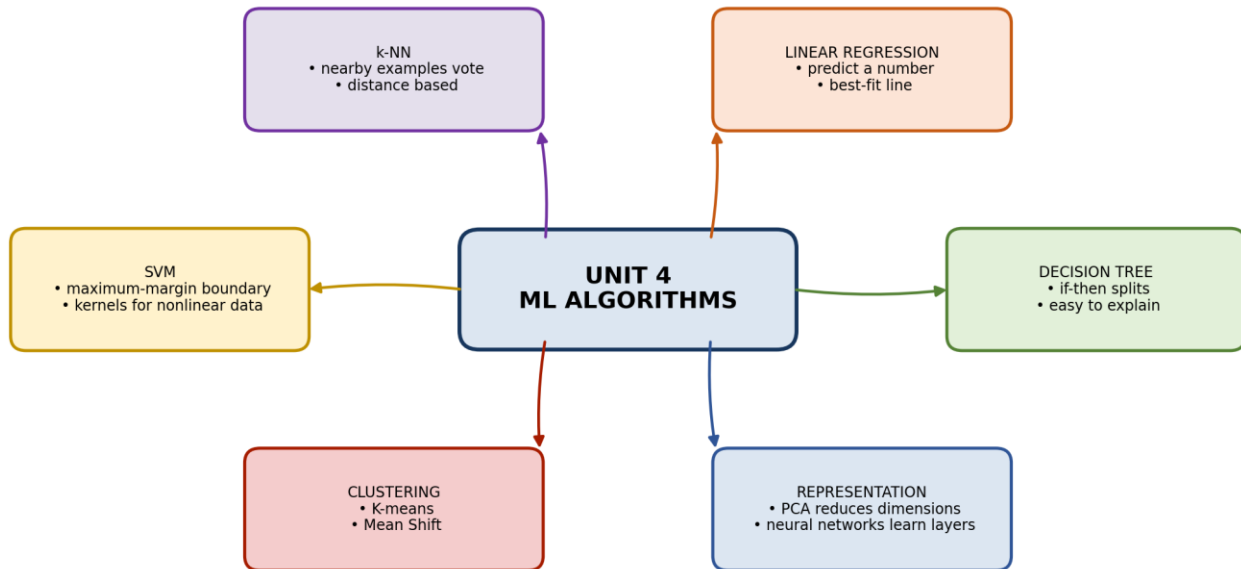
THE DATA SCIENCE PROCESS IS ITERATIVE, NOT A ONE-WAY PIPELINE



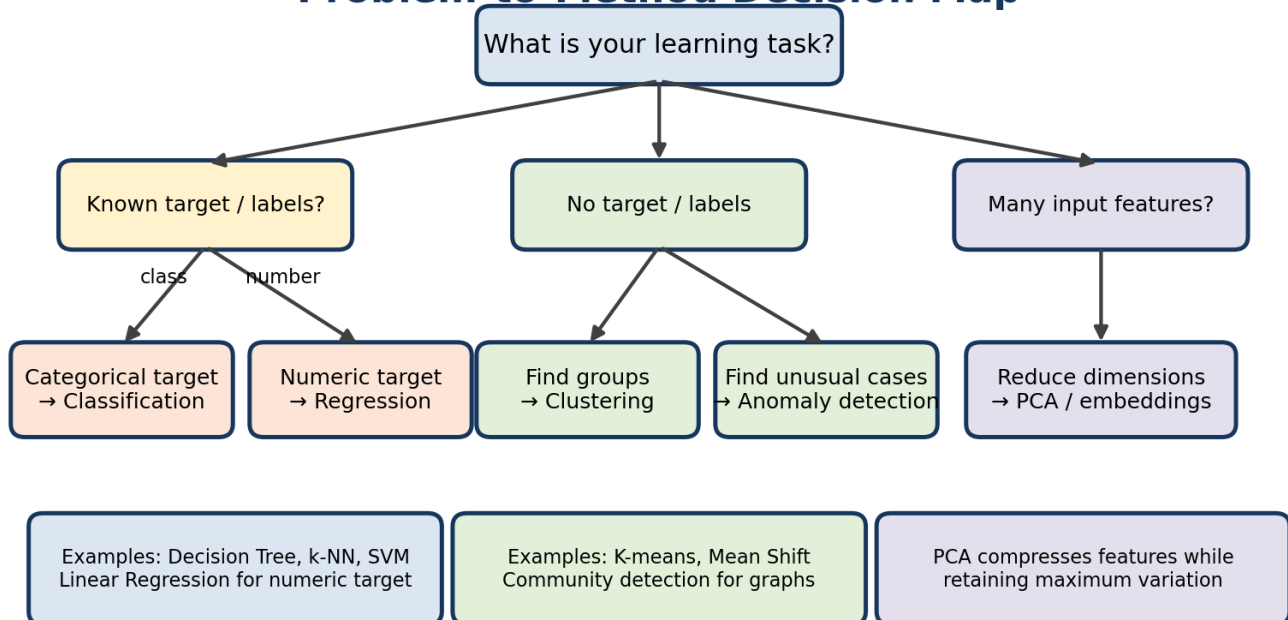
The process is iterative: findings often send us back to the question or data-preparation stage.

UNIT 4 — Machine Learning Algorithms

Unit 4 Mind Map



Problem-to-Method Decision Map

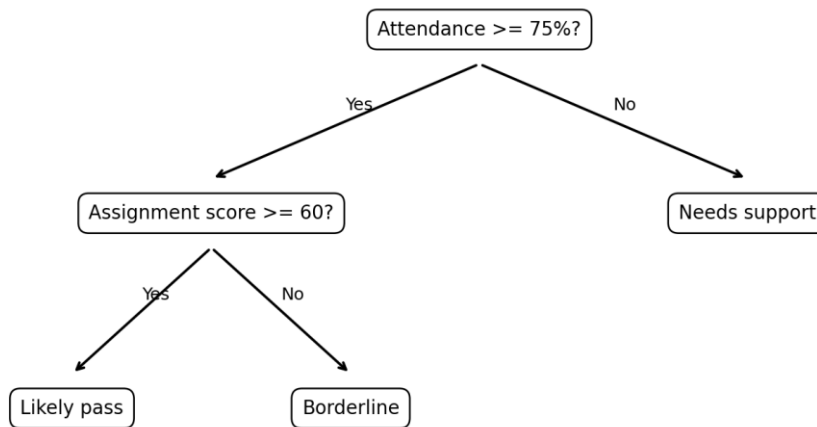


1. Supervised Algorithms

Algorithm	Core picture	Good for	Important setting
Decision Tree	A sequence of if-then questions	Explainable classification/regression	Depth, split criterion
Linear Regression	A best-fit line or plane	Predicting continuous numbers	Regularization, features
k-NN	Nearby examples vote or average	Simple local patterns	k and distance metric
SVM	A boundary with maximum margin	High-dimensional classification	C, kernel, gamma

Decision Tree: Learn by Splitting

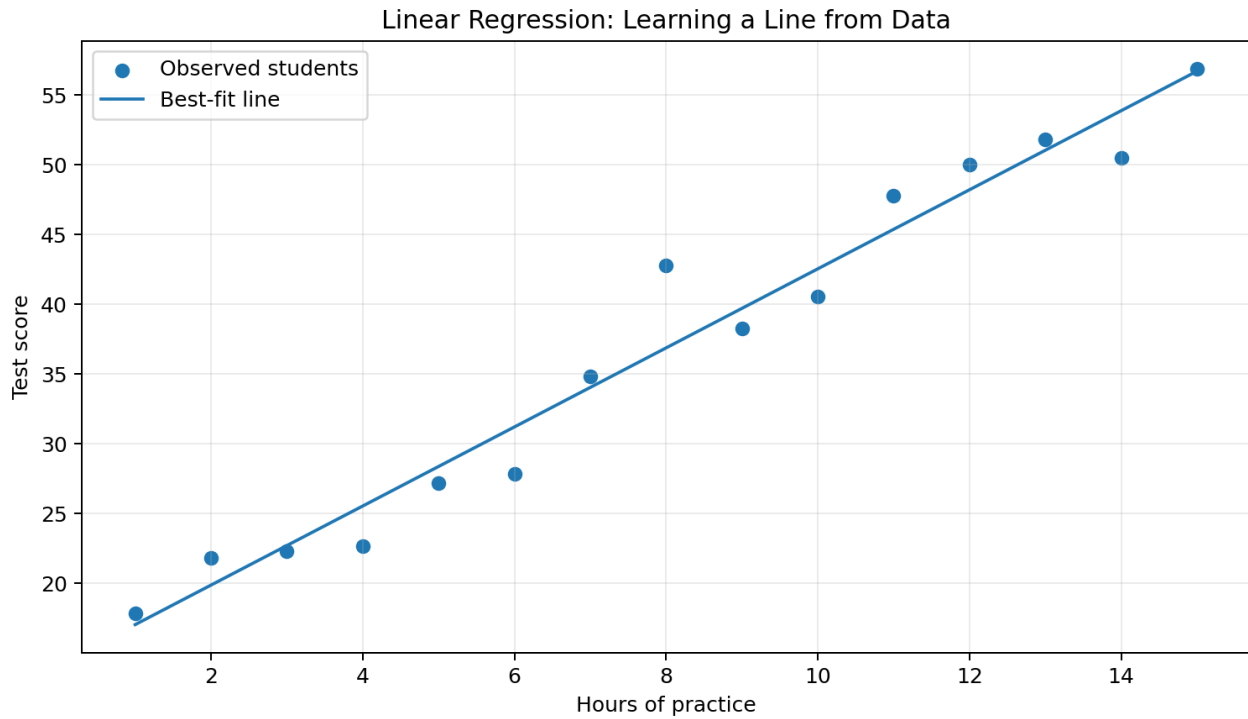
A Small Decision Tree for Student Support



HOW IT LEARNS: Intuition

At each node, choose the question that creates purer child groups. Stop before the tree memorises every training example.

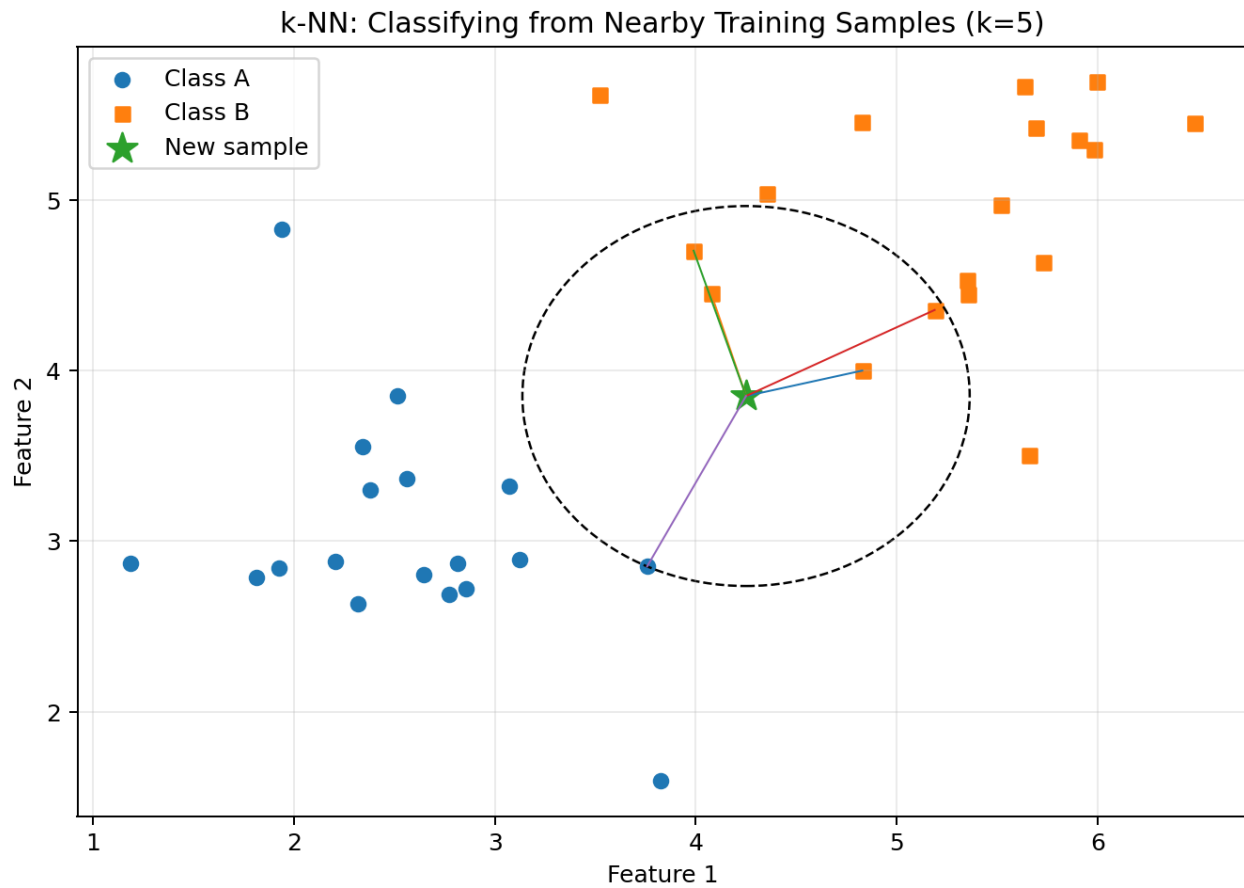
Linear Regression: Learn a Numeric Relationship



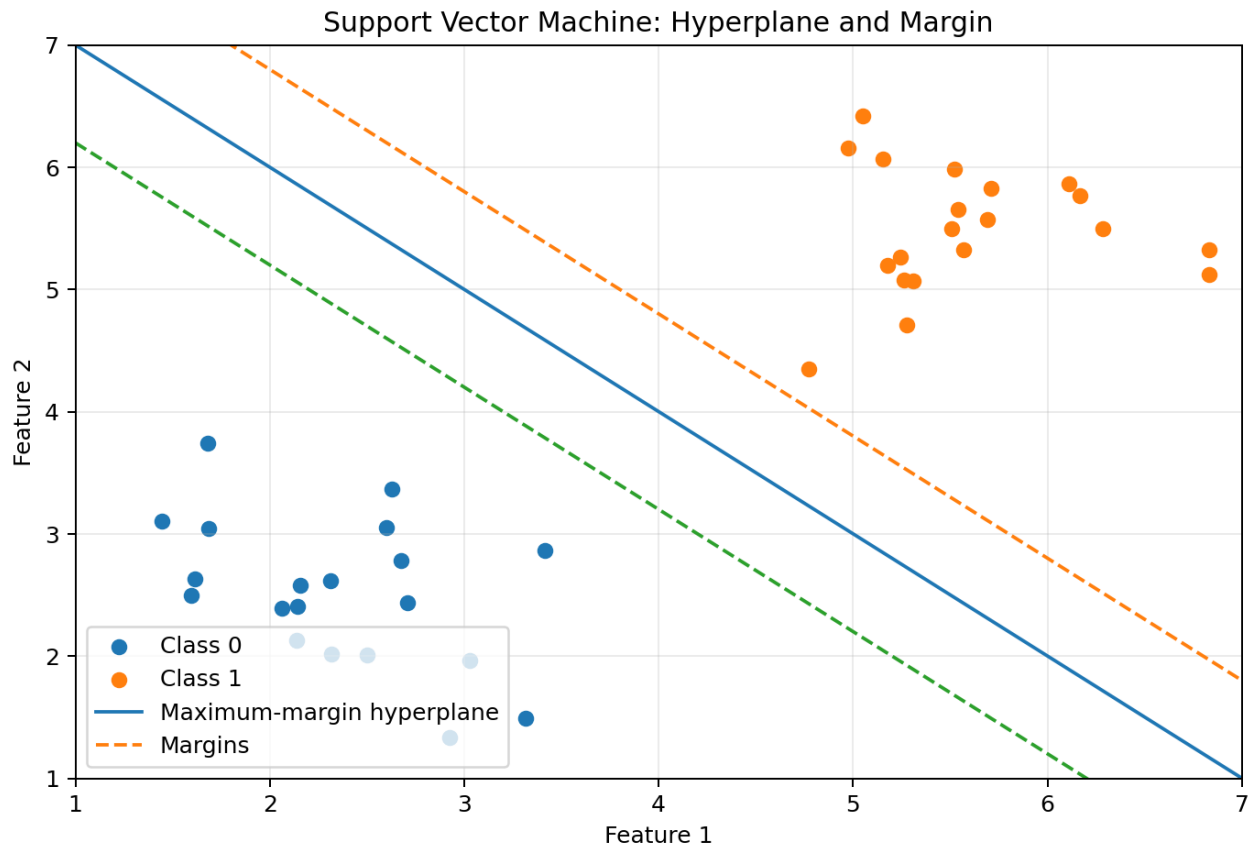
FORMULA IDEA: Residual

Residual = actual value – predicted value. Training selects coefficients that make the squared residuals as small as possible.

k-NN: Learn from Neighbours



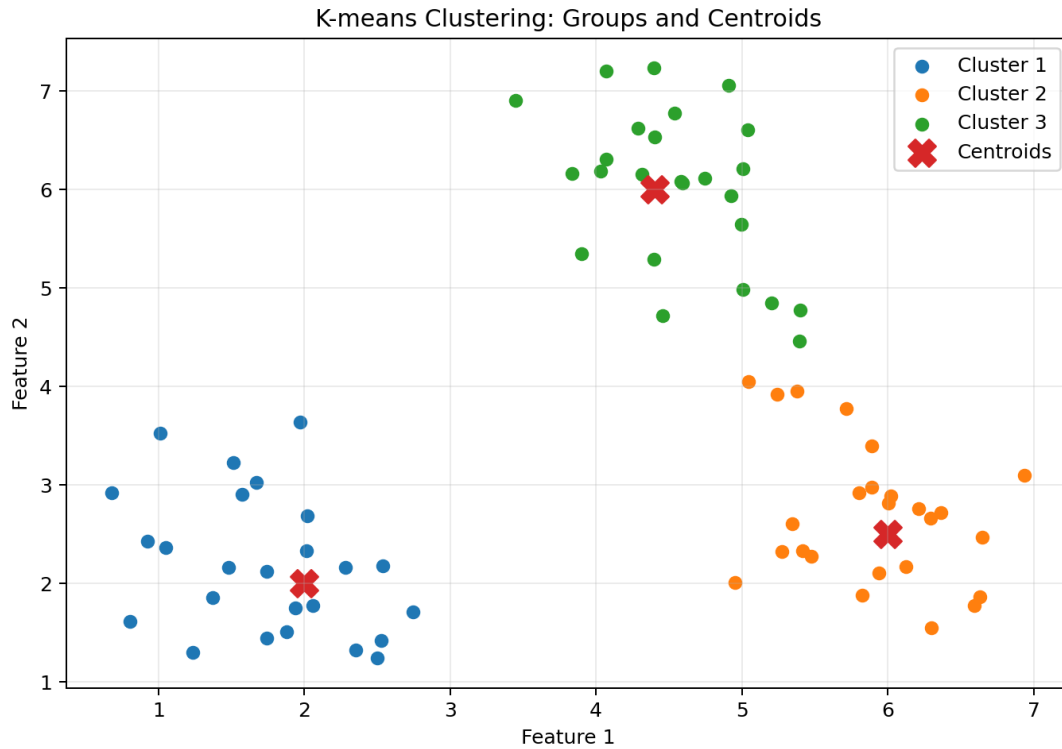
- Choose k , calculate distance from the new point to all training points, select the k nearest, then vote (classification) or average (regression).
- Small k can be noisy; very large k can hide local structure.
- Feature scaling is essential because large-unit features can dominate distance.

SVM: Maximum-Margin Boundary**KEY IDEA: Support vectors**

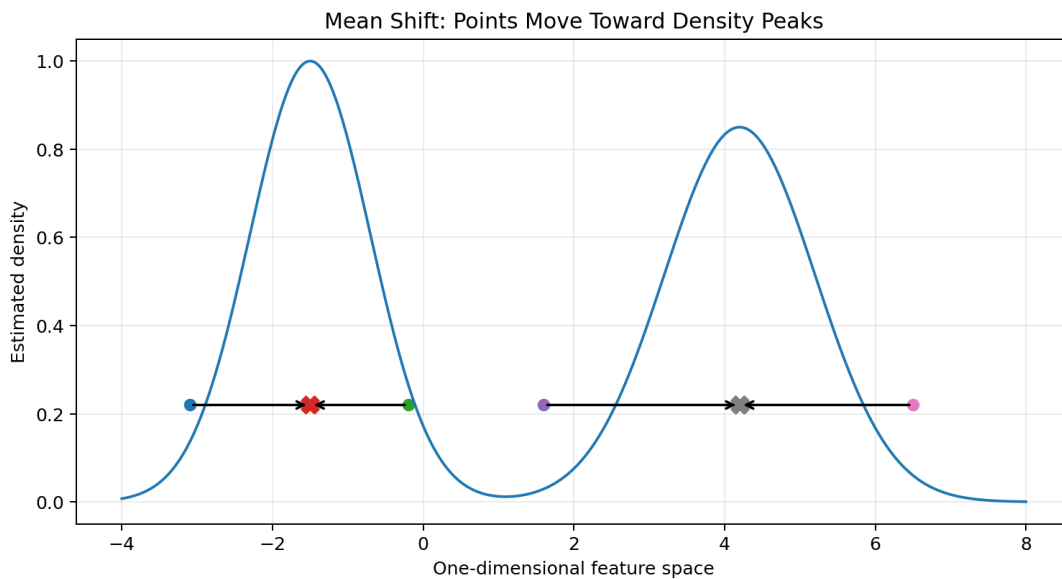
Only the training cases closest to the boundary directly determine the margin. A kernel allows a nonlinear boundary without manually constructing every transformed feature.

2. Unsupervised Algorithms

K-means: Group Around Centroids



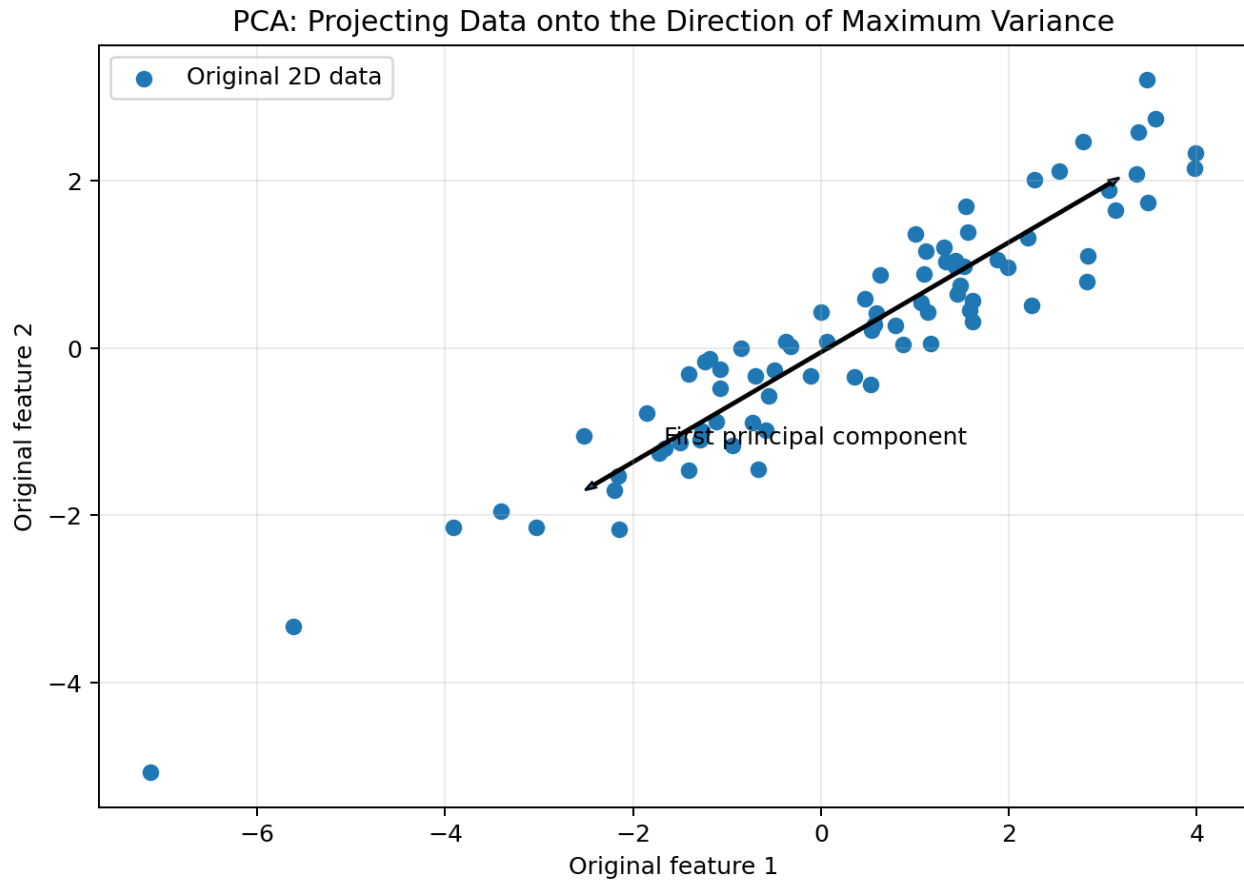
Mean Shift: Move Toward Dense Regions



K-means	Mean Shift
Must choose number of clusters k	Can discover number of modes/clusters
Fast and widely used	More computationally demanding
Works best for compact roughly spherical groups	Can follow irregular density peaks
Sensitive to initialization and scaling	Sensitive to bandwidth choice

3. Dimensionality Reduction and Neural Networks

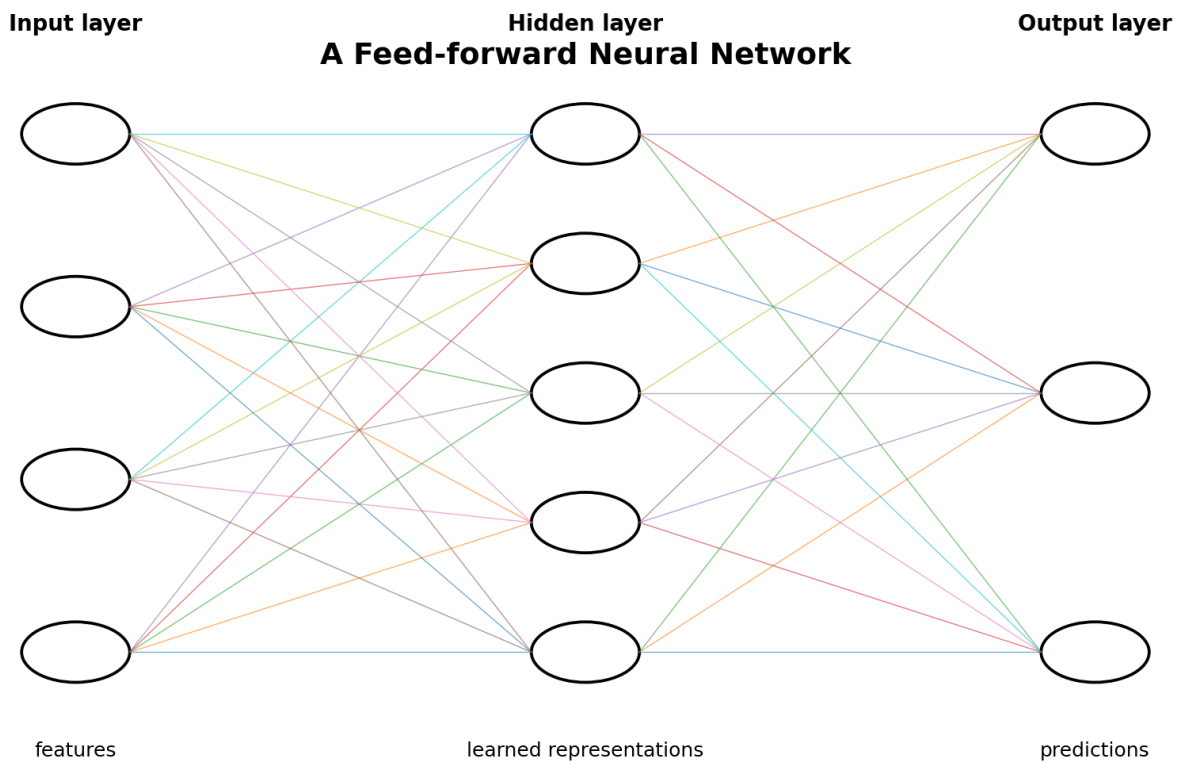
PCA: Compress Features



MENTAL MODEL: PCA picture

PCA rotates the coordinate system and keeps the directions that explain the most variation. It is unsupervised because it does not use the target label.

Neural Network: Layers of Weighted Units



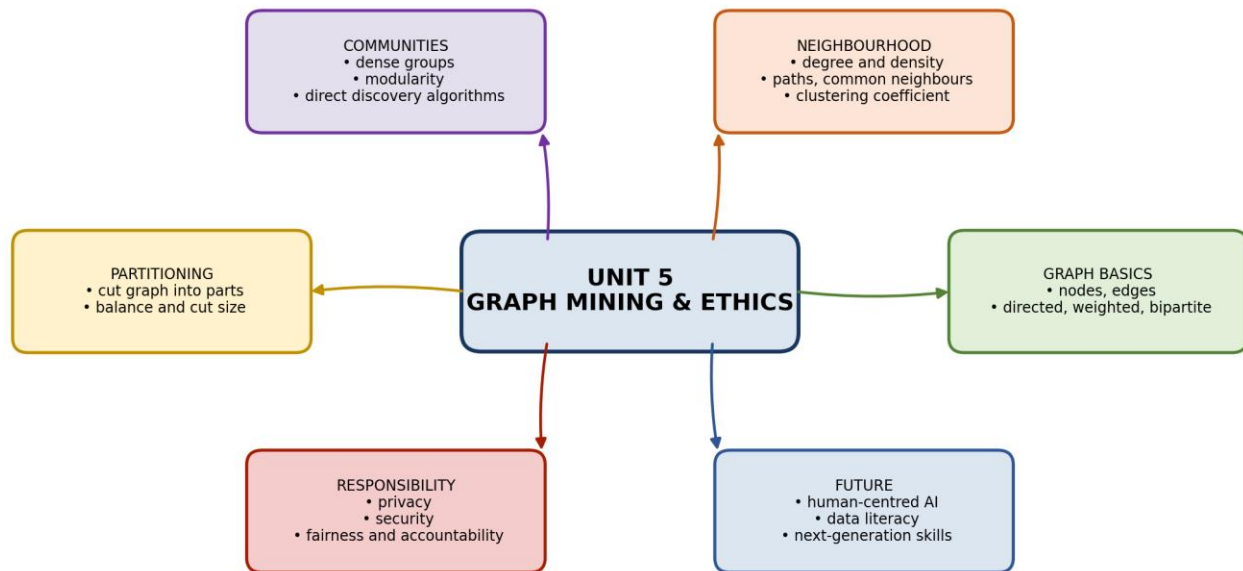
- **Forward pass:** Inputs move through weighted sums and activation functions to produce an output.
- **Loss:** The model measures how different the output is from the correct answer.
- **Backpropagation:** The error signal moves backward to calculate how each weight affected the loss.
- **Update:** An optimizer adjusts weights to reduce future loss.

DO NOT OVERSTATE: Common confusion

A neural network does not “think” like a human. It learns numerical transformations that are useful for reducing a training objective.

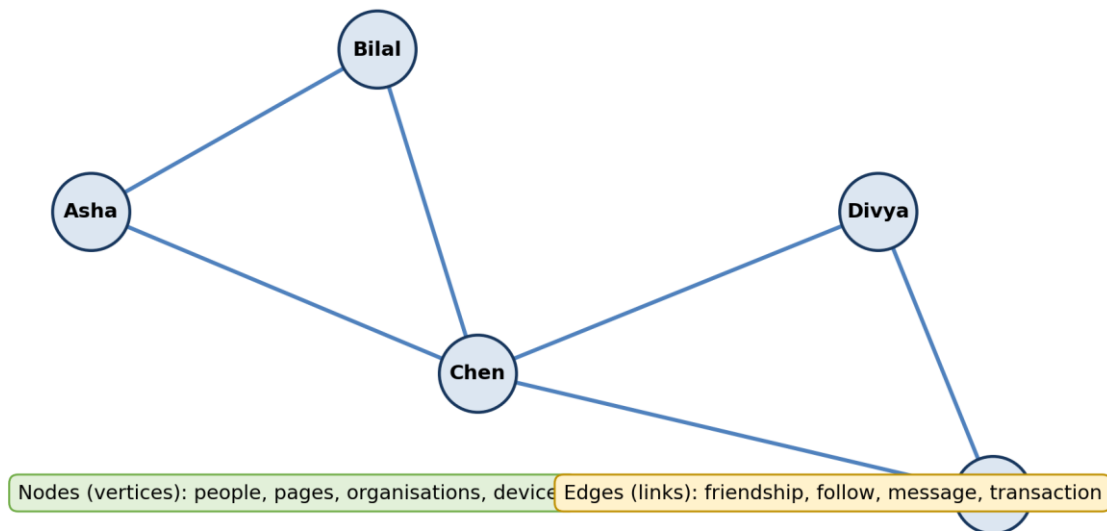
UNIT 5 — Mining Social-Network Graphs and Ethical Issues

Unit 5 Mind Map



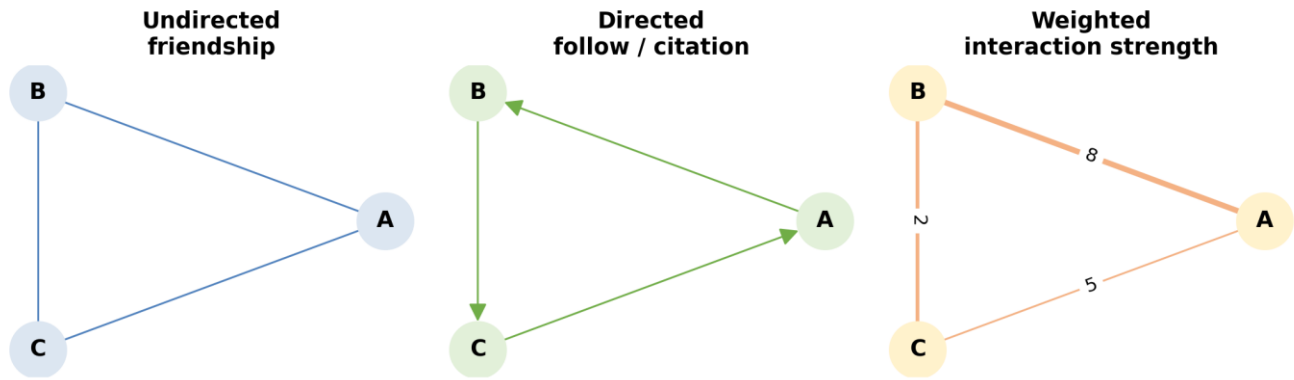
1. Social Networks as Graphs

A social network represented as a graph $G = (V, E)$



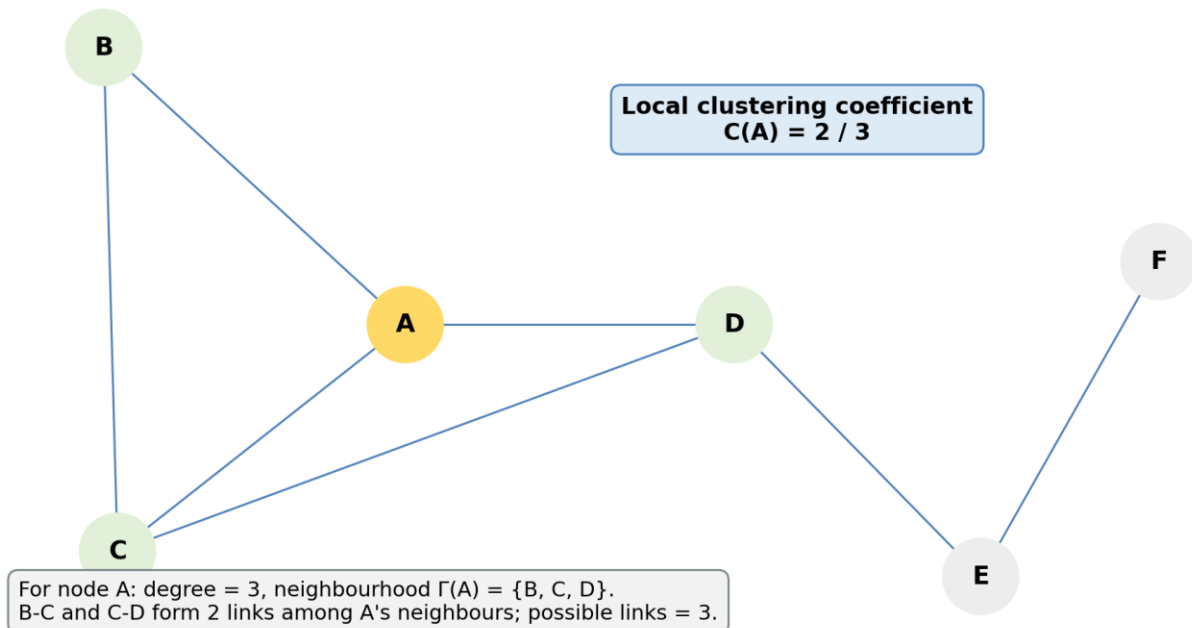
Graph element	Social-network meaning	Other examples
Node / vertex	Person or account	Webpage, product, city
Edge / link	Friendship, follow or message	Road, citation, transaction
Direction	Who follows or sends to whom	$A \rightarrow B$ may differ from $B \rightarrow A$
Weight	Strength or frequency of relationship	Number of messages or similarity

Common graph types used in social-network analysis



2. Neighbourhood Properties

Neighbourhood properties: degree, neighbours and triangles

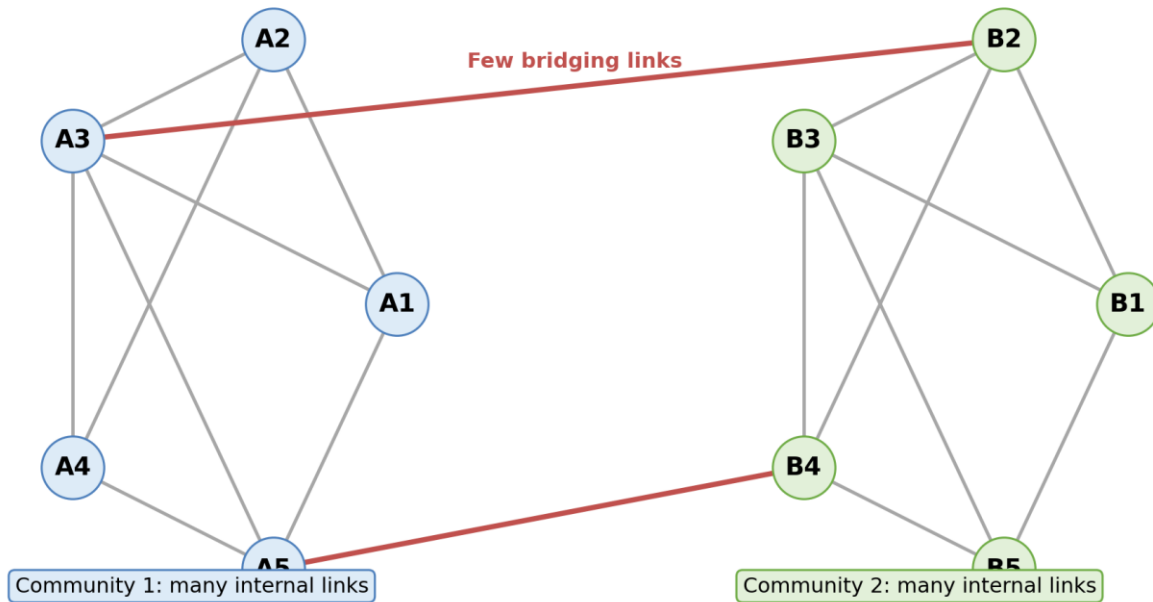


CONNECTION: Local versus global

Degree and clustering coefficient describe a node's nearby structure; components, paths, communities and partitions describe larger graph structure.

3. Communities, Clustering and Partitioning

Community structure: dense within groups, sparse between groups



Direct community discovery algorithms

Greedy modularity
Merge communities when modularity improves

Louvain / Leiden idea
Local moves + repeated aggregation

Girvan-Newman
Remove high-betweenness bridge edges

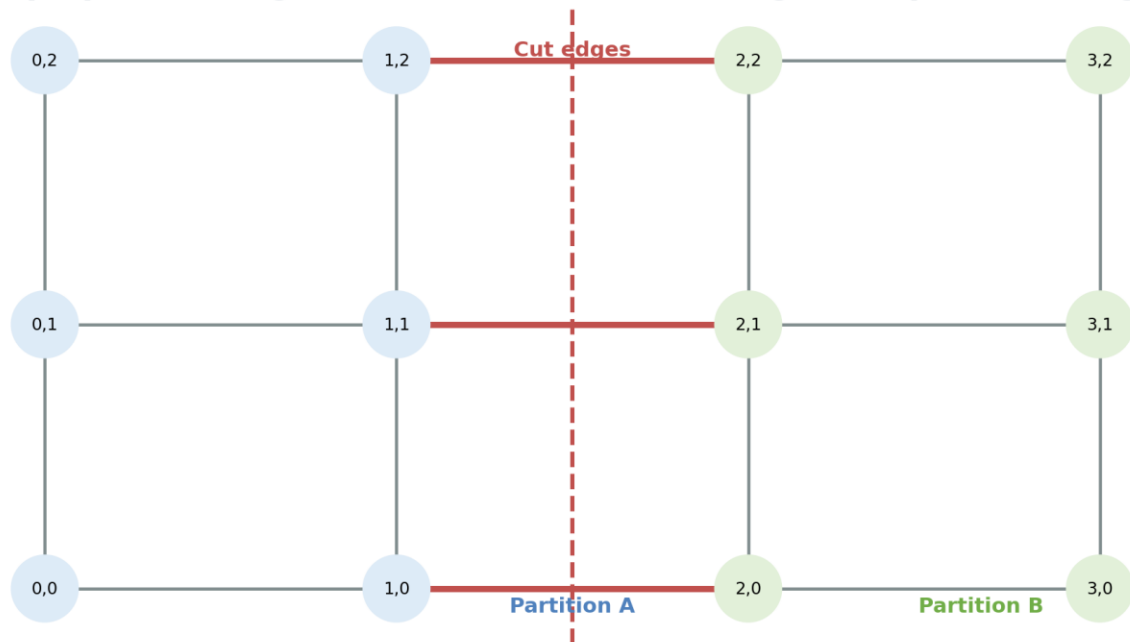
Label propagation
Nodes adopt common neighbour labels

Spectral methods
Use graph Laplacian eigenvectors

Clique / overlap methods
Allow nodes to belong to overlapping groups

No single method is universally best: choose based on graph size, direction, weights, overlap and interpretability.

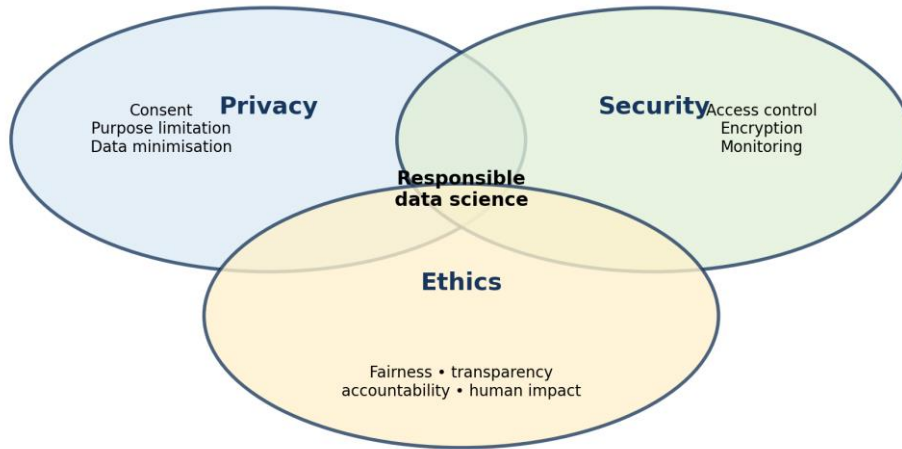
Method idea	How it discovers groups
Greedy modularity	Repeatedly merges groups that improve modularity
Label propagation	Nodes adopt labels common among neighbours
Girvan-Newman	Removes bridge-like edges with high betweenness
Louvain concept	Optimises local community moves and then aggregates communities

Graph partitioning: divide nodes while limiting cross-partition edges**COMPARE: Community detection versus partitioning**

Community detection seeks naturally dense groups. Partitioning usually divides the graph into a requested number or size of parts while minimising connections cut.

4. Privacy, Security and Ethics

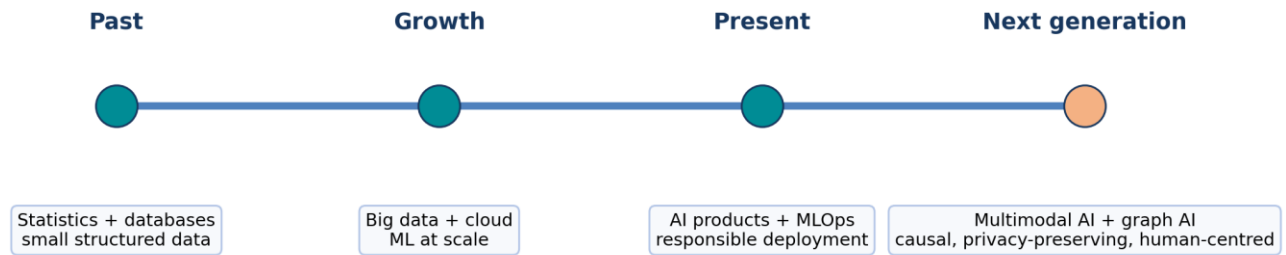
Privacy, security and ethics overlap but answer different questions



Question	Responsible practice
Do we need this data?	Collect only what is necessary for a clear purpose.
Can people be identified?	Remove direct identifiers and assess re-identification risk.
Who can access it?	Use access control, encryption and audit logs.
Could outcomes harm a group?	Test fairness, document limitations and provide human review.
Can the decision be challenged?	Provide explanation, correction and appeal mechanisms.

5. A Look Back and the Next-Generation Data Scientist

A look back at data science and the next generation of practitioners



Technical capability Statistics, programming, databases, machine learning, visualization and cloud tools.	Problem capability Translate a real need into measurable questions and useful decisions.
Human capability Communicate with non-technical people, collaborate and understand context.	Responsible capability Protect privacy, reduce harm, test fairness and monitor deployed systems.
CAREER IDEA: Future-ready mindset The next-generation data scientist is not only a model builder. They are an evidence-based problem solver who understands people, systems, uncertainty and responsibility.	

Integrated Concept Connections

Concept A	Connects to	Why the connection matters
Scaling	k-NN, SVM, K-means, PCA	All depend on distances or feature magnitude.
Sampling	Inference and model evaluation	A biased sample produces biased conclusions and models.
EDA	Preprocessing and algorithm choice	Plots reveal missing values, outliers, class imbalance and nonlinearity.
Distance metrics	k-NN, clustering, graph similarity	The meaning of “near” controls the result.
Train/test split	Performance measures	Evaluation must use unseen data.
Graph communities	Clustering	Both search for groups, but graphs use connections rather than only feature coordinates.
Ethics	Every stage	Harm can enter during collection, modelling, deployment or interpretation.

Ten Questions to Ask Before Using Any Model

1. What real decision or problem are we trying to improve?
2. Who is represented in the data, and who may be missing?
3. What is one row, what are the features, and what is the target?
4. Are labels correct, lawful and meaningful?
5. Which preprocessing steps are required, and were they fitted only on training data?

6. Is the task classification, regression, clustering, dimensionality reduction or graph analysis?
7. Which metric matches the cost of errors?
8. Does performance remain good for important subgroups and new time periods?
9. Can the result be explained and challenged?
10. How will the model be monitored, updated or stopped?

One-Page Formula and Memory Sheet

Topic	Formula / memory rule
Standardization	$z = (x - \text{mean}) / \text{standard deviation}$
Min-max scaling	$x' = (x - \text{minimum}) / (\text{maximum} - \text{minimum})$
Euclidean distance	Square root of the sum of squared coordinate differences
Manhattan distance	Sum of absolute coordinate differences
Accuracy	Correct predictions / all predictions
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$
F1-score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$
Regression residual	Actual – predicted
MSE	Average of squared residuals
Node degree	Number of edges incident on a node
Jaccard similarity	Intersection size / union size
Supervised learning	Learns with target labels
Unsupervised learning	Finds structure without target labels

Common Confusions — Corrected

Confusion	Correct understanding
AI and ML are identical	ML is one important approach within the wider field of AI.
Normalization always means 0 to 1	Min-max scaling maps features to a range; vector normalization changes each sample length.
A high accuracy always means a good model	Accuracy can be misleading with imbalanced classes or unequal error costs.
Correlation proves causation	Correlation shows association, not necessarily cause.
Clustering labels are correct answers	Clusters are algorithm-created groupings that require interpretation.
PCA selects the best original features	PCA creates new components as combinations of original features.
A graph community must be fully connected	A community is relatively denser internally, not necessarily a complete clique.
Removing names guarantees privacy	People may still be re-identified through combinations of attributes or graph structure.

Rapid Self-Test

1. Differentiate Data Science, Data Mining and Machine Learning using one example.
2. Why should a scaler be fitted on training data rather than the full dataset?
3. Choose a plot for one numeric variable and a plot for two numeric variables.
4. Differentiate a population parameter from a sample statistic.
5. When would a non-parametric method be preferred?
6. Why does k-NN require scaling?
7. Differentiate classification, regression, clustering and dimensionality reduction.
8. What is the role of support vectors in SVM?
9. Differentiate graph community detection and graph partitioning.
10. Give one privacy risk and one fairness risk in a social-network analysis project.

Suggested Visual Revision Routine

Day	Visual focus	Active task
1	Master map + Unit 1	Explain AI–DS–ML relationship aloud.
2	Unit 2	Draw preprocessing workflow from memory.
3	Unit 3	Create your own population–sample example.
4	Unit 4 supervised	Compare tree, regression, k-NN and SVM.
5	Unit 4 unsupervised	Explain K-means, Mean Shift, PCA and neural networks.
6	Unit 5	Draw a small graph and identify neighbours and communities.
7	Whole subject	Answer the ten model questions and rapid self-test.

SUBJECT IN ONE LINE: Final memory chain

Question → Data → Prepare → Explore → Learn → Evaluate → Communicate → Deploy → Monitor responsibly.